HIMANSHU WARUDKAR

AI/ML Engineer | Agentic AI | RAG & LLM Systems
+91-7020437634 | himanshuwarudkar9@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of production experience designing enterprise-grade AI systems, RAG pipelines, agentic workflows, and computer vision applications. Skilled in LangChain, CrewAI, MCP, Amazon Bedrock, YOLOv8/v11, and Azure cloud. Proven track record building scalable, compliance-safe, low-hallucination AI systems that deliver measurable business impact.

TECHNICAL SKILLS

Languages	Python, SQL
Gen AI / Agents	LangChain, LangGraph, CrewAI, AutoGen, MCP, RAG, Amazon Bedrock, Prompt Engineering, OpenAI API
ML/DL	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, PyCaret
Computer Vision	YOLOv8/v11, OpenCV, Torchreid, Re-ID, Object Detection & Tracking
Cloud & Data	Azure (Data Factory, Databricks, Synapse), AWS SQS, PySpark, Pinecone, ChromaDB, Vector Databases
Databases & Tools	PostgreSQL, MySQL, MongoDB, FastAPI, REST API, Git, Docker, Streamlit, Jira

WORK EXPERIENCE

Junior AI Developer | Leah, Mumbai, India — 06/2026 - Present

- Contributing to Leah, ContractPodAi's enterprise multi-agent AI orchestration platform — fixing bugs across the Vue 3/TypeScript frontend, Python/FastAPI backend, and flow-editor microservice.
- Resolved issues in real-time agent-run streaming (LangGraph SDK, SSE/WebSockets), multi-database persistence (MongoDB, PostgreSQL, Qdrant), and async task processing (Redis/Celery).
- Worked with a multi-LLM agent stack (deepagents SDK, LangChain/LangGraph, Claude, OpenAI, Gemini) to fix integration and workflow bugs in production.

Associate Data Science Engineer II | Techdome, Indore, India — 07/2025 - 06/2026

Worldpay Enterprise Email Automation

- Architected an enterprise RAG pipeline for a global payments environment: emails processed for intent/keyword extraction → KB article retrieval → extractive compression (article ranking, semantic filtering, query-focused filtering) to minimize token usage and hallucination risk before LLM response generation.
- Deployed Amazon Bedrock (Claude Haiku) with enterprise prompts enforcing strict KB grounding, escalation handling, and compliance-safe behavior; system generates customer-ready email responses and internal CSM notes with focus on auditability and deterministic outputs.

RiseAngle instant Video - AI Video Generation Pipeline

- Engineered an end-to-end multi-model AI pipeline (GPT-4→ Leonardo/Kling AI→ ElevenLabs → FFmpeg) converting content ideas into fully produced short-form videos; built an episodic content system maintaining consistent AI-generated characters and story arcs across multiple episodes.
- Designed a dual-queue AWS SQS architecture separating real-time user requests from background pre-generation workers, masking multi-model latency to deliver near-instant video results; integrated HeyGen avatar generation as an alternative async content pipeline on FastAPI.

Data Scientist | Squad Sync, Hyderabad, India — 05/2024 - 06/2025

- Developed contactless biometric system using rPPG signal processing, achieving 97% accuracy in face-based vitals and 98% in heart rate detection from video.
- Engineered multi-sensor physiological pipeline (video + audio), cutting biometric data processing time by 25%; ensured GDPR-compliant data integration and presented system refinements at IIIT Hyderabad CIE Labs Workshop on Computer Vision (2025).

PROJECTS

AI Equity Trader using MCP GitHub↗ | Python FastMCP pydantic-ai SQLite Yahoo Finance. Gradio

- Architected an autonomous multi-agent trading system for NSE/BSE markets with 3 independent LLM-powered trader agents coordinated via a Trading Floor orchestrator, each managing its own portfolio and executing decisions every 60 minutes via async/await concurrency.
- H Built 3 FastMCP servers (Accounts, Market, Search) as the agent tool layer - handling buy/sell execution, real-time Yahoo Finance price fetching with SQLite caching (<100ms/operation), and Serper web research - with a live Gradio dashboard refreshing portfolio at 5s and agent logs at 2s intervals.

Medical AI Hub - RAG-Powered Clinical Assistant GitHub↗ | LangChain Pinecone Gemini 2.0 Flash Hugging Face. Flask.gTTS

- Built a multimodal clinical Q&A system over a medical textbook using RAG (500-token chunks, all-MiniLM-L6-v2 384-dim embeddings, top-3 cosine retrieval from Pinecone serverless) achieving 1.5-2s text and 3-5s multimodal query latency with Gemini Vision for medical image analysis.
- Engineered a hybrid prompt pipeline fusing Gemini Vision findings, Google Speech-to-Text (WebM→WAV via pydub), and RAG chunks into a single grounded call; gTTS voice output with LLM temperature 0.3 to minimise hallucinations in clinical responses.

EDUCATION & CERTIFICATIONS

PG Diploma - Big Data Analytics
CDAC, Bengaluru | 73.88% | 08/2023

B.E. Mechanical Engineering
GH Raisoni College, Nagpur | CGPA: 8.23 | 11/2018

- Microsoft Certified: Azure AI Fundamentals (AI-900), 2025
- Machine Learning Rock Star, Infosys Springboard, 2024
- IIIT Hyderabad CIE Labs - Computer Vision Workshop, 2025